

Ejercicio 1: Gestión de Notas

Enunciado:

Crea una aplicación que permita:

1. Listar todas las notas.
2. Añadir una nueva nota (un texto simple).
3. Eliminar una nota seleccionada.

Estructura del proyecto:

```
/mvc-notes
  /controllers
    NoteController.php
  /models
    NoteModel.php
  /views
    notes.php
    index.php
```

Solución:

Modelo (NoteModel.php)

```
<?php
class NoteModel {
    private $notes = [];

    public function __construct() {
        $this->notes = [
            ['id' => 1, 'content' => 'Estudiar para el examen'],
            ['id' => 2, 'content' => 'Llamar al médico']
        ];
    }

    public function getNotes() {
        return $this->notes;
    }

    public function addNote($content) {
        $id = count($this->notes) + 1;
        $this->notes[] = ['id' => $id, 'content' => $content];
    }

    public function deleteNote($id) {
        $this->notes = array_filter($this->notes, function($note)
use ($id) {
            return $note['id'] != $id;
        });
    }
}
```

Controlador (NoteController.php)

```
<?php
require_once 'models/NoteModel.php';

class NoteController {
    private $model;

    public function __construct() {
        $this->model = new NoteModel();
    }

    public function listNotes() {
        $notes = $this->model->getNotes();
        require 'views/notes.php';
    }

    public function addNote($content) {
        $this->model->addNote($content);
        $this->listNotes();
    }

    public function deleteNote($id) {
        $this->model->deleteNote($id);
        $this->listNotes();
    }
}
```

Vista (views/notes.php)

```
<!DOCTYPE html>
<html>
<head>
    <title>Gestión de Notas</title>
</head>
<body>
    <h1>Notas</h1>
    <ul>
        <?php foreach ($notes as $note): ?>
            <li>
                <?php echo htmlspecialchars($note['content']); ?>
                <a href="?action=delete&id=<?php echo $note['id'];
?>">Eliminar</a>
            </li>
        <?php endforeach; ?>
    </ul>

    <h2>Añadir Nota</h2>
    <form method="post" action="?action=add">
        <input type="text" name="content" placeholder="Nueva nota"
required>
        <button type="submit">Añadir</button>
    </form>
</body>
</html>
```

Punto de entrada (index.php)

```
<?php
require_once 'controllers/NoteController.php';

$controller = new NoteController();

$action = $_GET['action'] ?? 'list';

switch ($action) {
    case 'add':
        $content = $_POST['content'] ?? '';
        if (!empty($content)) {
            $controller->addNote($content);
        }
        break;

    case 'delete':
        $id = $_GET['id'] ?? 0;
        if ($id) {
            $controller->deleteNote($id);
        }
        break;

    default:
        $controller->listNotes();
        break;
}
```

Ejercicio 2: Gestión de Productos con Descuento

Enunciado:

Crea una aplicación que permita:

1. Listar productos con sus precios originales y precios con descuento (10%).
2. Añadir un nuevo producto con su precio.

Estructura del proyecto:

```
/mvc-products
  /controllers
    ProductController.php
  /models
    ProductModel.php
  /views
    products.php
    index.php
```

Solución:

Modelo (ProductModel.php)

```
<?php
class ProductModel {
    private $products = [];

    public function __construct() {
        $this->products = [
            ['id' => 1, 'name' => 'Producto A', 'price' => 100],
            ['id' => 2, 'name' => 'Producto B', 'price' => 200]
        ];
    }

    public function getProducts() {
        return $this->products;
    }

    public function addProduct($name, $price) {
        $id = count($this->products) + 1;
        $this->products[] = ['id' => $id, 'name' => $name, 'price'
=> $price];
    }
}
```

Controlador (ProductController.php)

```
<?php
require_once 'models/ProductModel.php';

class ProductController {
    private $model;

    public function __construct() {
        $this->model = new ProductModel();
    }

    public function listProducts() {
        $products = $this->model->getProducts();
        require 'views/products.php';
    }

    public function addProduct($name, $price) {
        $this->model->addProduct($name, $price);
        $this->listProducts();
    }
}
```

Vista (views/products.php)

```
<!DOCTYPE html>
<html>
<head>
    <title>Gestión de Productos</title>
</head>
<body>
    <h1>Productos</h1>
    <table border="1">
        <tr>
            <th>Nombre</th>
            <th>Precio Original</th>
            <th>Precio con Descuento</th>
        </tr>
        <?php foreach ($products as $product): ?>
            <tr>
                <td><?php echo htmlspecialchars($product['name']);
?></td>
                <td><?php echo $product['price']; ?> €</td>
                <td><?php echo $product['price'] * 0.9; ?> €</td>
            </tr>
        <?php endforeach; ?>
    </table>

    <h2>Añadir Producto</h2>
    <form method="post" action="?action=add">
        <input type="text" name="name" placeholder="Nombre del
producto" required>
        <input type="number" name="price" placeholder="Precio"
required>
        <button type="submit">Añadir</button>
    </form>
</body>
</html>
```

Punto de entrada (index.php)

```
<?php
require_once 'controllers/ProductController.php';

$controller = new ProductController();

$action = $_GET['action'] ?? 'list';

switch ($action) {
    case 'add':
        $name = $_POST['name'] ?? '';
        $price = $_POST['price'] ?? 0;
        if (!empty($name) && $price > 0) {
            $controller->addProduct($name, $price);
        }
        break;

    default:
        $controller->listProducts();
        break;
}
```